



What do you want to play?



Staying in the Queue:

Which Spotify Users Are Most Likely to Leave — and How to Keep Them

BANA 205 – Fundamentals of Marketing | Fall 2025

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NYSE: **SPOT**

Public in **2018**

Spotify is a Swedish music streaming and media services company founded by Daniel Ek and Martin Lorentzon on April 23, 2006.

Revenue Model

Freemium Business Model

Ad-Supported	Premium
Display, audio & video advertising	Ad-free subscription
★ Display, audio & video advertising delivered through advertising impressions ★ Arrangements with advertising agencies ★ Accounted for 13% of revenue Spotify made in 2024	★ A music listening experience without commercial breaks. ★ Unlimited online and offline high-quality streaming access to Spotify's catalog of music and podcasts ★ Accounted for 87% of revenue Spotify made in 2024

Target Customers

Listeners

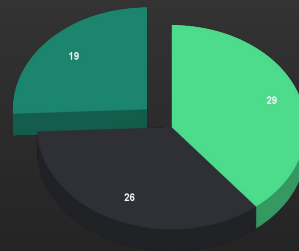
Spotify's global audience across free and premium tiers

Advertisers

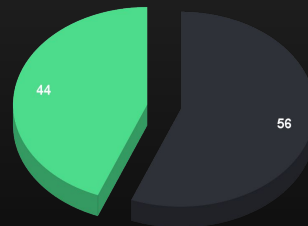
- ★ Brands
- ★ Businesses
- ★ Advertising agencies
- ★ Educational institutions
- ★ Spotify Ad Studio companies

Content Creators

Podcasters, bands and artists distributing original content through the platform



• Millennials • > Age 24 • < Age 55+



• Male • Female

“It all comes back to user fundamentals and that’s where we are: 700 million users who keep coming back, engagement at all-time highs. We’re building Spotify for the long-term.”

– Daniel Ek, Spotify Founder and CEO



Statistical Highlights			
2025 MONTHLY ACTIVE USERS 713M CHANGE 11.33%	2025 PREMIUM SUBSCRIBERS 281M CHANGE 11.51%	2025 MARKET CAPITALIZATION \$120.44B CHANGE 34.60%	2025 TOTAL REVENUE €4.27B CHANGE 7.15%
2024 PREMIUM ARPU €4.71 CHANGE -3.68%	2025 PREMIUM ARPU €4.53 CHANGE -3.82%	2025 OPERATING MARGIN 2.20% CHANGE -1.20%pt	2025 AD-SUPPORTED REVENUE €446M CHANGE -5.91%

Sources: Spotify Q3 2025 Earnings Release (Published Nov 2025)
Spotify Q3 2024 Shareholder Deck (Released Nov 2024)
Spotify Q3 2025 Shareholder Deck / 6-K Filing (Released Nov 2025)
Spotify Share Price & Market Data — YFinance (Accessed Nov 2025)

The Marketing Problem

—————→ Rising Churn Threatens Spotify's Growth



Why This Matters

- Spotify's freemium model depends on user retention
- Premium subscriptions generate 87% of revenue in 2024
- Industry competition (Apple Music, YouTube Music) makes churn costly

Analysis Question

Which types of Spotify users are most at risk of churn, and how can marketing tactics be tailored to improve retention?

Marketing Process Focus

Segmentation

Segment users based on behavior and churn risk

Targeting

Identify and target highest-risk clusters

Positioning

Personalize campaigns that reinforce loyalty

Retention

Increased retention & reduced churn

Marketing Strategy "STP"

About Our Data

Spotify Customer Churn dataset

- ★ A Spotify-style user dataset (1,000 users) sourced from Kaggle
- ★ Includes basic demographics, listening behavior, subscription type, and churn status

The dataset tells us:

- Who users are
- How they listen
- User behavior
- User status

Limitations

- The dataset is synthetic.
- Results illustrate approach and insights, not real numbers.
- This protects user privacy while still allowing meaningful analysis.



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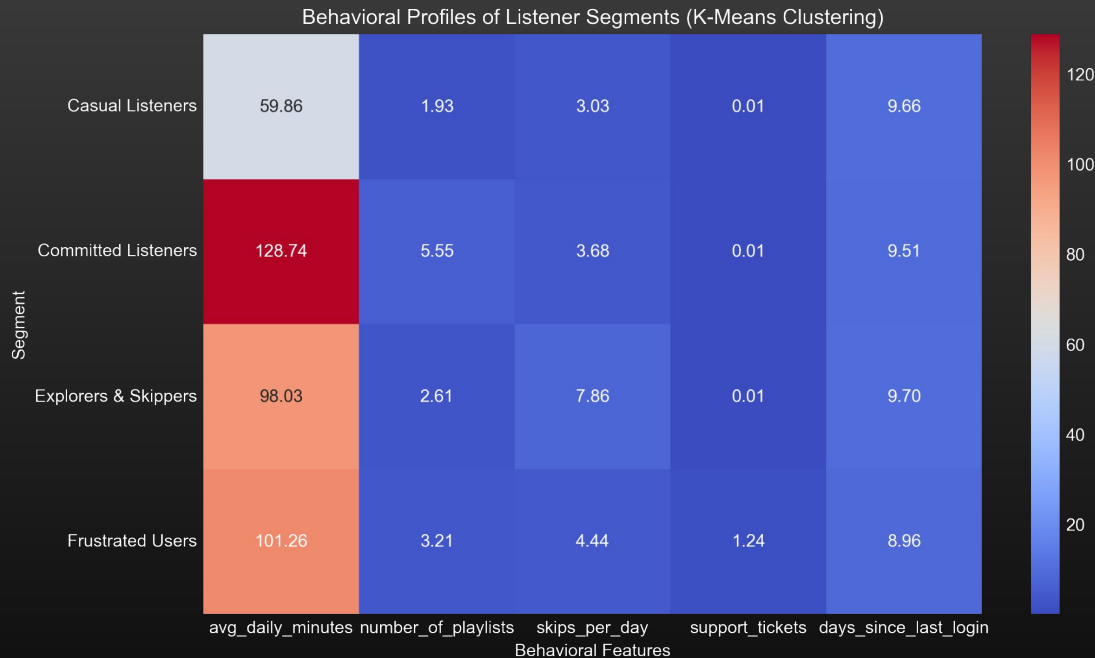
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Analysis I

Which users are at risk of churning?

Identifying At-Risk Users

- ★ Used K means clustering on selected behavioral variables
- ★ Elbow method → selected $k = 4$ groups
- ★ Uncovered behavioral spotify listener profiles and linked them to churn patterns



Analysis I – K-means Clustering

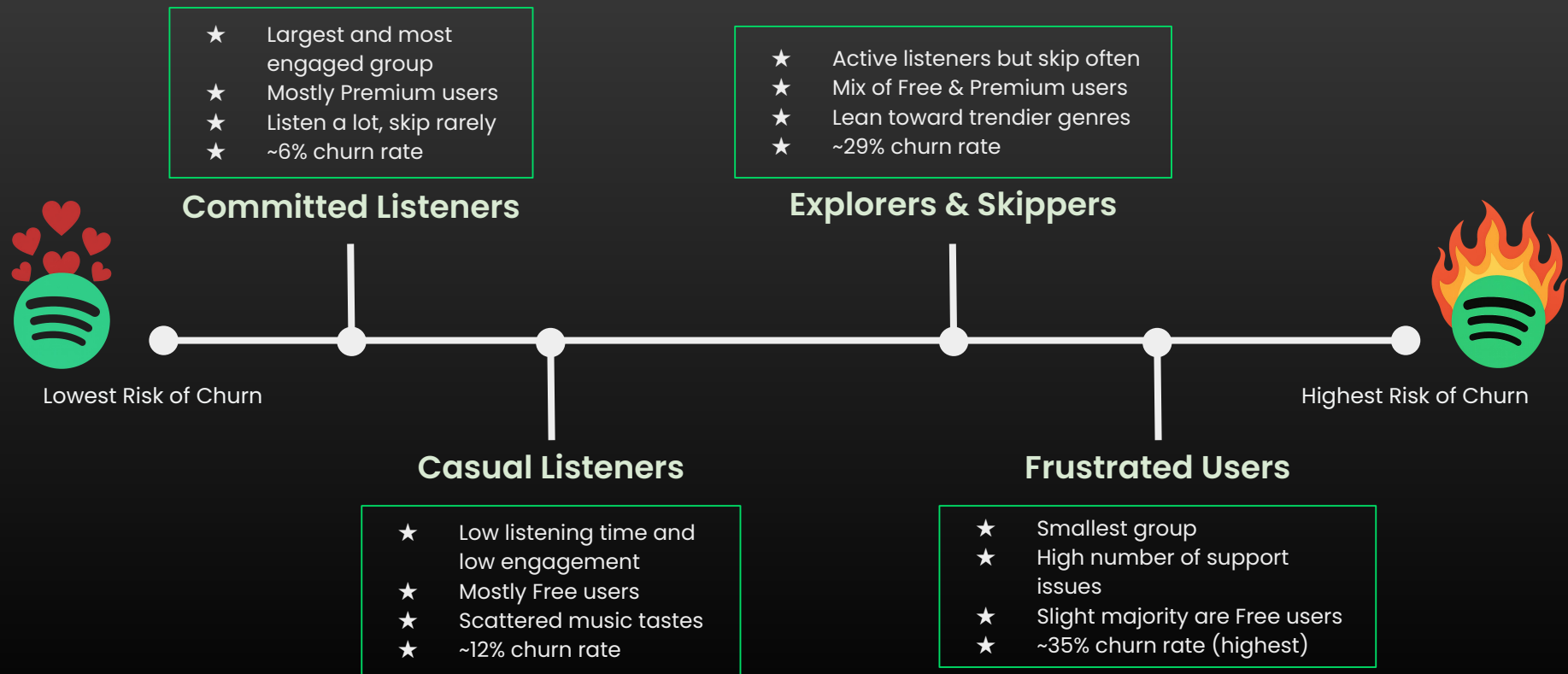
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Churn Likelihood by Cluster



Analysis I

Limitations and Robustness

Limitations

- Dataset is synthetic → patterns appear cleaner & more predictable
- Limited behavioral depth (no session data, no device info, no ad exposure)
- No Premium subtypes (e.g., Family, Duo, Student, Individual)
- Cluster boundaries & churn differences may be exaggerated due to simplified data

Robustness Check

- ★ Tested different cluster counts ($k=3, 4, 5$) → overall patterns similar; $k=4$ chosen
- ★ Removed key behavioral variables individually → segment structure remained broadly consistent
- ★ Results show **acceptable stability** given the synthetic dataset



Analysis I – K-means Clustering
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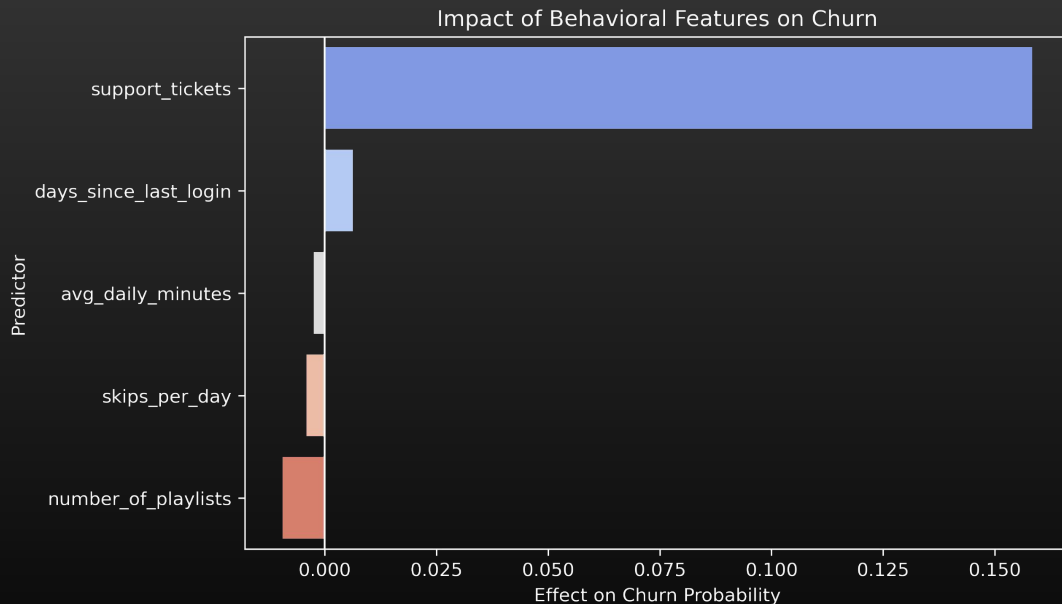


Analysis II

What specific behaviors are stronger predictors of churn?

We ran a multilinear regression using:

- ★ Target variable:
Churn (0 = stayed, 1 = churned)
- ★ Predictors:
 - Support tickets
 - Days since last login
 - Avg daily minutes
 - Skips per day
 - Number of playlists



Analysis II – Multilinear Regression

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Analysis II

Key Findings

Support tickets are the strongest churn predictor

Users contacting support are much more likely to churn

Inactivity increases churn

More days since last login = higher churn risk

Engagement helps retention

More listening time & playlists slightly reduce churn

Skipping behavior has a small positive effect on churn

Not as significant as support tickets or inactivity



Analysis II – Multilinear Regression

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Analysis II

Limitations, Robustness and Model Fit

Model Fit	Multilinear Regression Model Fit explains 15.3% of variation in turn ($R^2 = 0.153$) • Normal and expected for behavioral churn data • Human churn behavior is complex and is explained beyond behavioral attributes.
Robustness Check	★ Results stay consistent when removing individual predictors.
Limitations	• The model only uses behavioral variables—no demographics, no device type, no subscription age, etc. • Relationships are weaker and less realistic due to the synthetic dataset.



Analysis II – Multilinear Regression

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Overall Results

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#	Title	Findings
1	High-Churn Segments	○ Frustrated Users ■ High skip rates and poor listening experience ■ Highest churn rate (~35%) ○ Explorers & Skippers ■ Active but skip frequently ■ High churn rate (~29%)
2	High-Churn Drivers	○ Engagement → Less time results in higher churn ○ Skips → More skips indicates lower satisfaction ○ Subscription → Free users churn more ○ Content Fit → Poor match results in higher churn



Spotify Churn Analysis Overall Results

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Managerial Implications

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#	Title	Point
1	Prioritize High-Risk Segments With Targeted Retention Strategies	○ Skippers: Low engagement and weak listening habits ■ Context-Based playlist prompts (commute, study, workout) ○ Frustrated Listeners: Negative experiences with ads or irrelevant content ■ Short term Premium trials (7-14 days) ■ Lighter ad loads to reduce friction on free users
2	Use Segmentation to Guide Personalized Messaging	○ Different listener types respond to different motivations
3	Leverage Multi-Format Content to Build Stickier Listening Habits	○ High-churn users tend to consume only one type of audio content ○ Encourage multi-format engagement to deepen loyalty



Managerial Implications

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Marketing Recommendations

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#	Title	Strategy
1	Retain & Reward Committed Users	○ Introduce loyalty perks: early album access & exclusive sessions ○ Gamify listening streaks: badges, shareable stats, playlist milestones ○ Promote premium add-ons: family upgrade incentives
2	Personalize Content to Each Segment	○ Recommendation engine tuned differently per listener type ○ Promote artists and genres that reduce skipping behavior ○ Use behavioral triggers
3	Improve Experience for Frustrated Users	○ Offer short Premium Trial windows after support tickets ○ Reduce ad-load for users with high skip rates



Marketing Recommendations
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What do you want to play?



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Thank You!

1,345,762 monthly listeners



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Discography

Show all

1	Segmented users based on behavior and churn risk	22,349,725	4:20
2	Identified high-risk clusters & identified features that contribute to churn	24,143,346	3:55
3	Offered recommendations to increase retention & reduce churn	14,339,885	4:26



Thank You



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